

Technical Document #853: Retrieval Augmented Generation - Comprehensive Overview

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Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- Hybrid search combines keyword-based and semantic search.
- Query decomposition breaks complex queries into simpler sub-queries.
- Evaluation metrics for RAG include faithfulness, answer relevance, and context recall.